

MERN Stack Development – Module 3 Exercise

Hands-on Exercise No. 3

Course: MERN Stack Development

Module: Express

Total Marks: 10

Due Date: 17/05/2026

Instructions:

- Prepare your solution in MS Word
 - Submit in .doc or .docx format
 - Late or copied submissions will not be accepted
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Learning Outcome:

- Implement routing in Express
 - Use middleware for request handling
 - Apply authentication in APIs
 - Handle errors properly
 - Design RESTful APIs
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Problem Statement:

You are working as a Backend Developer for a startup building a student portal. The client requires a secure and scalable API to manage student records efficiently.

Task Requirements:

- Create a new Express project
- Implement the following routes:
 - GET /students → Fetch all students
 - POST /students → Add a new student
 - GET /students/:id → Fetch a student by ID
- Add middleware for logging incoming requests
- Implement authentication middleware (simple token-based system)

- Add error handling for:
 - Invalid routes
 - Missing or incorrect data
 - Test your API using Postman
 - Take screenshots of:
 - API requests
 - API responses
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Submission Guidelines:

- Upload your .docx file on LMS
- Include:
 - Code snippets
 - Postman screenshots
 - Output responses
- Add short explanation for each feature